

Exercise 1 Solution

1.

a)

1) FCFS

P1	P2	P3	P4	P5	
0	10	11	13	14	19

2) nonpreemptive SJF

P1	P2	P4	P3	P5	
0	10	11	12	14	19

3) preemptive SJF

P1	P2	P3	P4	P5	P1	
0	2	3	5	6	11	19

4) nonpreemptive priority

P1	P2	P5	P3	P4	
0	10	11	16	18	19

5) preemptive priority

P1	P2	P1	P5	P1	P3	P4	
0	2	3	6	11	16	18	19

或

P1	P2	P1	P5	P3	P1	P4	
0	2	3	6	11	13	18	19

6) RR (quantum = 1)

We assume that when a preempted process and new coming process need to be put in the ready queue at the same time, we prefer to put the new coming process in front of the preempted process in the ready queue.

P1	P1	P2	P1	P3	P4	P1	P3	P5	P1	P5	P1	P5	P1	P5	P1	P5	P1	P1
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0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19

7) RR (quantum = 3)

P1	P2	P3	P1	P4	P5	P1	P5	P1	
0	3	4	6	9	10	13	16	18	19

b)

1) $[0 + (10-2) + (11-3) + (13-4) + (14-6)]/5 = 6.6 \text{ ms}$

2) 6.4 ms

3) 2 ms

4) 8 ms

5) 6.6 ms / 6 ms

6) $[(19-10) + 0 + (8-3-2) + (6-4-1) + (17-6-5)]/5 = 3.8 \text{ ms}$

7) $[(19-10) + (4-2-1) + (6-3-2) + (10-4-1) + (18-6-5)]/5 = 4.6 \text{ ms}$

c)

1) $[10 + (11-2) + (13-3) + (14-4) + (19-6)]/5 = 10.4 \text{ ms}$

2) 10.2ms

3) 5.8 ms

4) 11.8 ms

5) 10.4 ms / 9.8 ms

6) $[(19-0)+(3-2)+(8-3)+(6-4)+(17-6)]/5 = 7.6$ ms

7) $[(19-0)+(4-2)+(6-3)+(10-4)+(18-6)]/5 = 8.4$ ms

2.

a)

P1	P1	P2	P1	P2	P3	P4	P2	P5	P4	P2	P5	P4	
0	8	12	20	25	28	36	44	46	54	70	82	92	100

b) **Context switch:** The number of context switch is 8 (12 is also considered to be correct).

c) **average waiting time:**

$[(25-17)+(82-12-25)+0+(100-36-32)+(92-46-18)]/5=22.6$ ms

average turnaround time:

$[(25-0)+(82-12)+(36-28)+(100-36)+(92-46)]/5=42.6$ ms

3.

a)

P1	P1	P2	P1	P3	P4	P3	P5	P2	P1	P2	P4	
0	8	10	18	28	36	44	52	60	76	84	90	94

P3	P5	P1		P4	P5	P1	
94	98	108	112	119	123	131	135

b)

Average waiting time = 52.4 ms

Average turnaround time = 91.8 ms

4.

(b) SJF and (d) Priority can cause starvation!